

Superconducting films

X21 (2021) — English translation

In thin films In – O at low temperatures (of the order of 1 K), a superconductor-insulator phase transition controlled by an external magnetic field is observed (the transition occurs when the external magnetic field B changes).

Part A. Critical parameters (6.8 points)

The table “Experimental data” presents measurements of the dependences $R(T)$, where R is the resistance of the sample, T is the temperature, at different external magnetic fields B . Resistance measurement error $\varepsilon(R) = 0.5\%$. The graph (see Fig.) shows approximate types of dependencies $R(T)$ for some values B indicated in the table. Let us assume that the dependence of resistance $R(T, B)$ has the following form:

$$R(T, B) = R_C - \gamma T + A (B - B_C) T^\alpha,$$

где A, R_C, B_C — некоторые параметры, связанные со свойствами образца. Если на кривой $R(T)$ есть точка перегиба (то есть $\frac{\partial R}{\partial T} = 0$), то обозначим ее координаты, как (T_n, R_n) . Примечание: Для функции двух переменных аналогом производной выступает частная производная согласно определениям: [0] [1] Фактически, при вычислении частой производной по x мы полагаем y constant and vice versa, for example:

$$f(x, y) = 3x^2 + 2xy, \quad \frac{\partial}{\partial x} f(x, y) = 6x + 2y, \quad \frac{\partial}{\partial y} f(x, y) = 2x$$

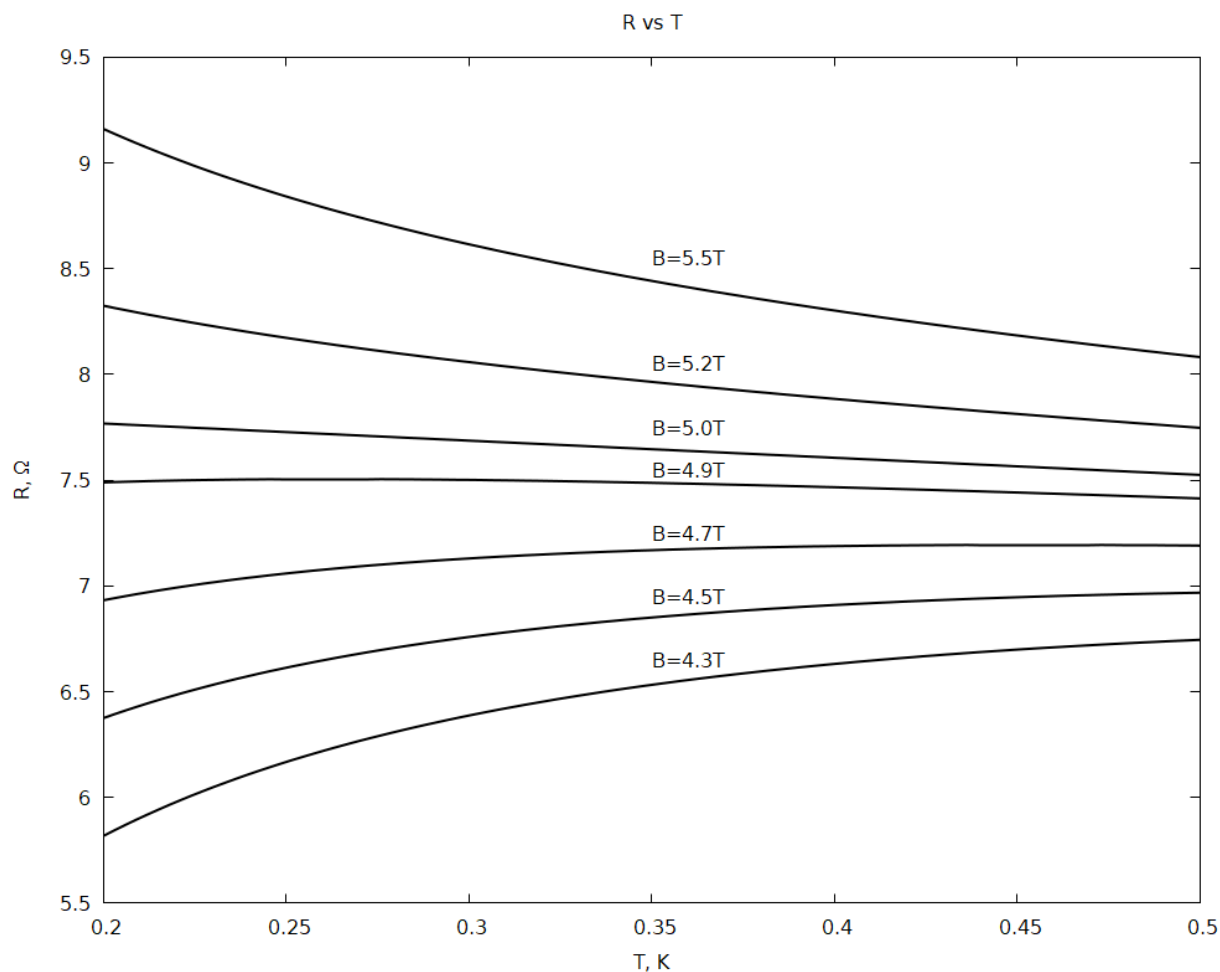


Fig. 1.

A1	Derive the analytical relationship $R_{\pi}(T_{\pi})$.	1.00
A2	Define B_C .	0.40
A3	Define R_C in two ways.	2.60
A4	Define the parameters γ and α .	2.80

Part B. Probability of describing an experiment (3.2 points)

Estimating the error of a physical quantity X allows one to calculate the probability with which its value lies in the interval $[a, b]$. To assess the probability p that the result of the experiment is described by a mathematical expression, the criterion χ^2 (chi-square) is used, calculated according to the following formula:

$$\chi^2 = \sum_{i=1}^N \left(\frac{R_{\text{теор}} - R_{\text{эксп}}}{\Delta R} \right)^2$$

For a number of points $N = 150$, a correspondence table p and χ^2 :

(Probability table)

B1	Count χ^2 (definition in note) of $R(T)$ dependencies for each B .	2.20
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B2 Using the table from the note, estimate the probability according to which the formula **1.00**
 $f = R(T, B)$ describes the dependence $R(T, B)$ for any (specify which) 150 points.

Source: <https://pho.rs/p/102>